

Supplementary Information 1

A registered genome-wide analysis identifies robust, context-dependent transcriptional responses of lychee to *Peronophythora litchii*

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Contents

Figures S1–S3, machine-readable Tables S1–S18, all figure source data, the registered protocol and amendment log, analysis code, workflows, configurations, tests, metadata, and environment specifications are available exclusively in the version-specific Zenodo record at <https://zenodo.org/records/22436625>. Their table and figure captions are listed below.

Supplementary table captions

The following captions correspond to the machine-readable TSV files archived in the Zenodo record.

- **Table S1.** Biological-unit registry.
- **Table S2.** Per-library quality control.
- **Table S3.** Genome-wide discovery statistics.
- **Table S4.** Robustness file manifest.
- **Table S5.** All external frozen tests.
- **Table S6.** Pathway and signature tests.
- **Table S7.** Conditional differential-transcript-usage results.
- **Table S8.** Candidate annotation and orthology evidence.
- **Table S9.** Small-RNA reference-gate outcome.
- **Table S10.** Motif-background tests and sensitivity results.
- **Table S11.** Accession-aware evidence registry.
- **Table S12.** Scripts, environments, and command inventory.
- **Table S13.** Protocol amendment and deviation log.

- **Table S14.** Legacy within-cultivar audit.
- **Table S15.** Controlled promoter-motif background comparison.
- **Table S16a.** Reconstructed dataset-search queries.
- **Table S16b.** Dataset eligibility decisions.
- **Table S17.** Exact software versions.
- **Table S18.** Power-simulation minimum detectable effects.

Supplementary figure captions

- Figure S1: replicate-level normalized counts.
- Figure S2: conditional parametric power curves.
- Figure S3: exploratory signed-signature estimate.

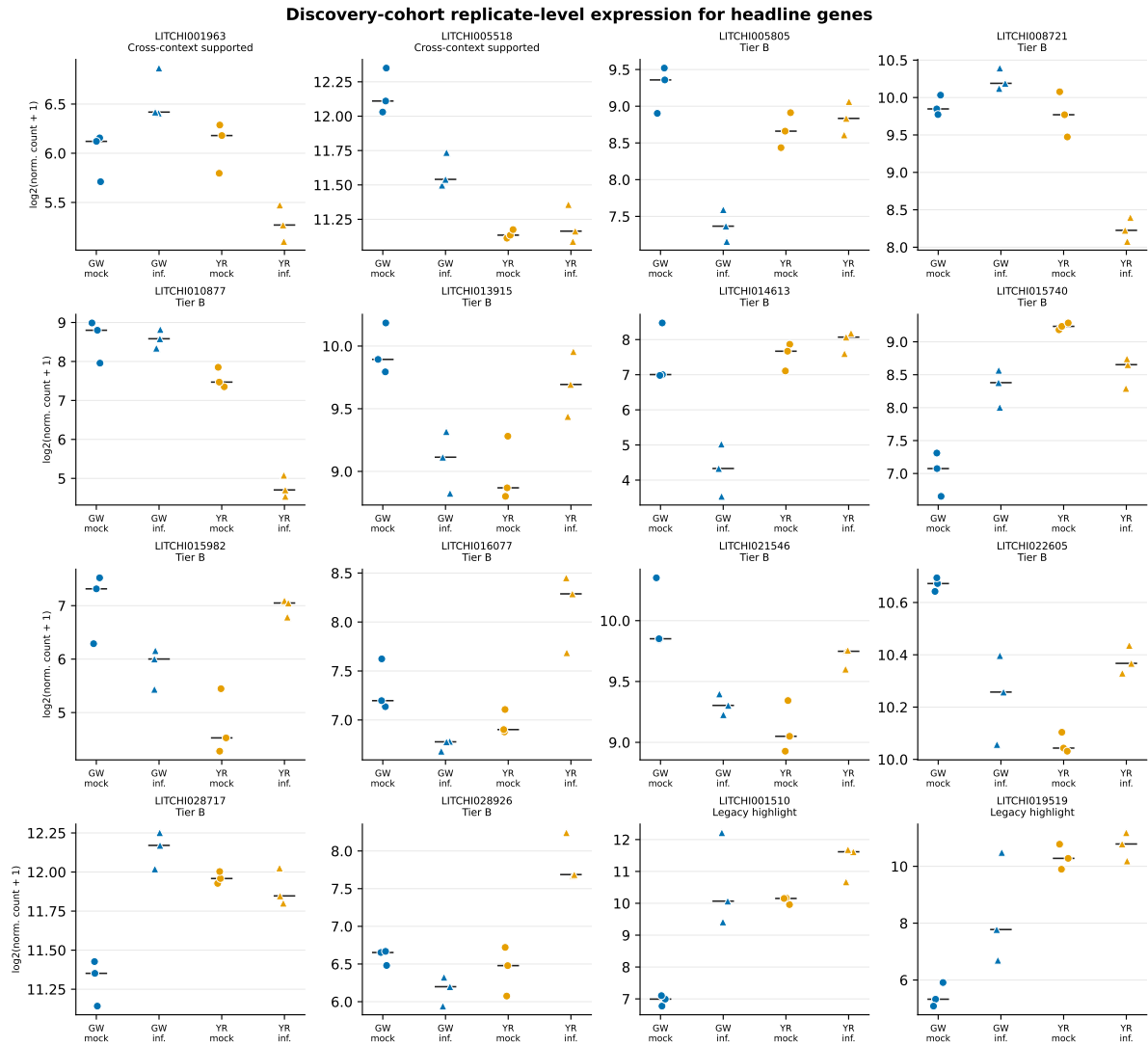
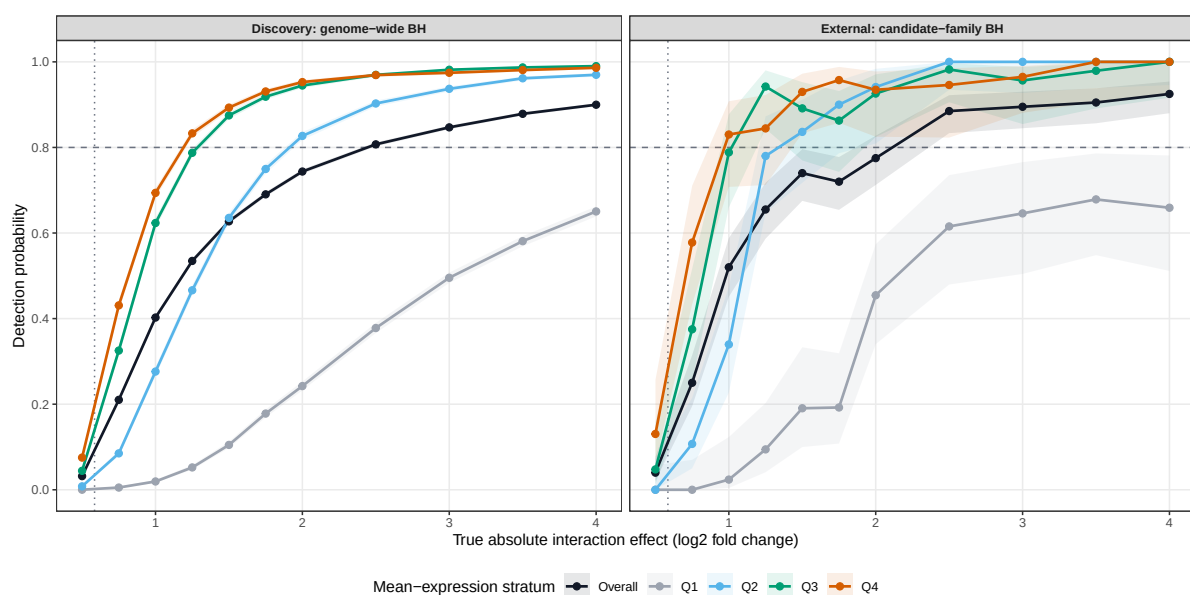


Figure S1: Replicate-level normalized counts for the two cross-context-supported genes, twelve Tier B genes, and two legacy highlights. Points are individual deposited libraries; bars show cultivar-treatment medians.



Dashed line: 80% power. Dotted line: registered $|\log_2\text{FC}|$ threshold. Parametric power is conditional on fitted means, dispersions, deposited-library independence, and the simulated non-null prevalence.

Figure S2: Parametric detection probability for cultivar-by-infection effects under genome-wide discovery and candidate-family external adjustment. Curves show the overall result and mean-expression quartiles; the dashed line marks 80% power.

S3 Exploratory signature estimate (not confirmatory)

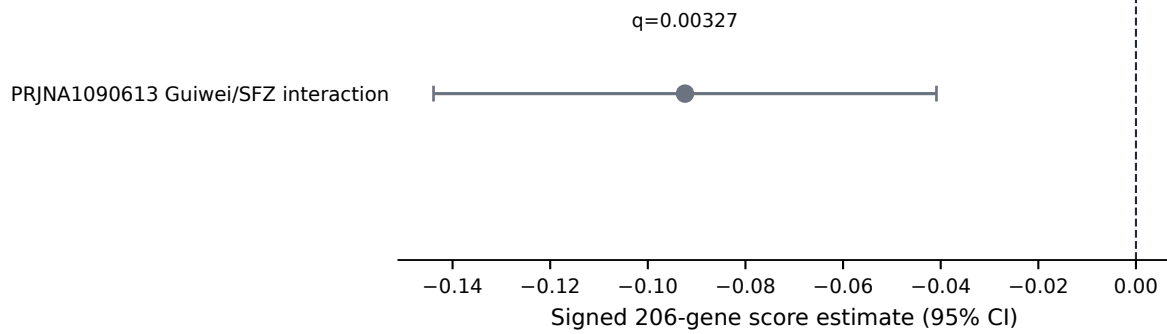


Figure S3: Exploratory PRJNA1090613 signed-signature estimate, shown separately because cohort metadata did not support its inclusion in the confirmatory evaluation.